

Anna, 35 · woke with **right facial droop including forehead** · cannot close right eye, taste reduced on anterior tongue · no other neuro findings · **Bell's palsy (CN VII)**

BIO 004 · WEEK 8 · CASE DEEP DIVE

The Cranial Nerves

Sudden right-sided facial droop including the forehead. A tear that won't close the eye. One peripheral nerve in trouble.

PATIENT CASE

Anna is 35. She woke this morning with the **right side of her face drooping**. She cannot close her right eye fully, cannot raise her right eyebrow, cannot smile on the right. Food tastes different on the front of her tongue. No weakness in her arm or leg; no slurred speech otherwise; no other deficits.

On exam she has **complete right-sided facial paralysis INCLUDING the forehead**, mild hyperacusis on the right (sounds louder), reduced taste on the anterior two-thirds of the right tongue. The diagnosis is **Bell's palsy** — an idiopathic peripheral palsy of cranial nerve VII (often viral-mediated inflammation).

The 12 cranial nerves come directly off the brain or brainstem (not the spinal cord) and serve the structures of the head and neck (plus some thoracic/abdominal viscera via CN X). Anna's palsy is one nerve in isolation, but every finding maps to a specific branch of CN VII.

GUIDING QUESTION

What do each of the 12 cranial nerves do, and how does a single peripheral CN VII lesion produce ipsilateral facial paralysis, loss of taste, and hyperacusis at the same time?

OOOTTAFGVAH

Classic mnemonic for names: *"On Old Olympus's Towering Tops A Finn And German Viewed Some Hops"* → Olfactory, Optic, Oculomotor, Trochlear, Trigeminal, Abducens, Facial, Vestibulocochlear, Glossopharyngeal, Vagus, (Spinal) Accessory, Hypoglossal. For function (sensory/motor/both): *"Some Say Marry Money But My Brother Says Big Brains Matter More"* (S = sensory, M = motor, B = both).

Function in one line each

#	NAME	TYPE	KEY FUNCTION
I	Olfactory	S	Smell. Passes through cribriform plate.
II	Optic	S	Vision. Optic canal.
III	Oculomotor	M	Most eye movements; eyelid; pupil constriction (parasympathetic).
IV	Trochlear	M	Superior oblique (downward + outward gaze).
V	Trigeminal	B	Sensation from face (3 divisions: V1 ophthalmic, V2 maxillary, V3 mandibular); motor to muscles of mastication.
VI	Abducens	M	Lateral rectus (lateral gaze).
VII	Facial	B	Muscles of facial expression; taste anterior 2/3 tongue; lacrimal/submandibular/sublingual glands; stapedius muscle.
VIII	Vestibulocochlear	S	Hearing + balance.
IX	Glossopharyngeal	B	Taste posterior 1/3 tongue; gag reflex (sensory); pharynx motor; parotid gland.
X	Vagus	B	Parasympathetic to most viscera; vocal cords; palate; sensation from larynx.
XI	Accessory	M	Sternocleidomastoid, trapezius.
XII	Hypoglossal	M	Tongue movement.

Why Anna has the combination of findings she has

Cranial nerve VII has multiple branches before it reaches the face. Damage at different points produces different combinations:

● Motor branches to facial muscles

Five branches at the parotid (temporal, zygomatic, buccal, mandibular, cervical). Innervate all muscles of facial expression.

● Chorda tympani

Carries taste from anterior 2/3 of tongue + parasympathetic to submandibular and sublingual glands.

● Greater petrosal nerve

Parasympathetic to lacrimal gland.

● Nerve to stapedius

Innervates the stapedius muscle that dampens loud sounds. Loss = hyperacusis (sounds too loud).

The forehead is the key

Why forehead involvement = peripheral (LMN) lesion. The forehead motor cortex sends fibers to BOTH facial nerve nuclei (bilateral cortical input). The lower face is controlled by only the CONTRALATERAL cortex. A cortical (UMN) stroke produces lower-face weakness on the contralateral side but spares the forehead (because the other hemisphere still drives forehead). A peripheral nerve (LMN) lesion takes out everything that nerve does — including the forehead. Anna can't wrinkle her right forehead → peripheral lesion, not a stroke.

Patterns worth memorizing

● CN III palsy

"Down and out" eye, ptosis, dilated pupil. Causes: posterior communicating artery aneurysm, diabetic microvascular ischemia.

● CN VI palsy

Cannot abduct eye → horizontal diplopia. Can be from raised intracranial pressure (long course).

● Trigeminal neuralgia

Lightning-like facial pain in V2 or V3 distribution. Often vascular compression.

● Acoustic neuroma

Tumor on CN VIII at the cerebellopontine angle. Hearing loss + tinnitus + balance issues. Can compress CN VII (facial weakness) and CN V (facial numbness).

Three special-sense, six motor, three mixed

Special senses: **I (smell), II (vision), VIII (hearing/balance)**. Pure motor: **III, IV, VI, XI, XII**. Plus IX (mostly mixed but sometimes considered motor-heavy). Mixed (sensory + motor): **V, VII, IX, X**.



DRAW ZONE 1 -- Sketch the base of the brain. Label all 12 cranial nerves emerging in order.

CN I from olfactory bulbs, II from optic chiasm, III-IV from midbrain, V-VIII from pons, IX-XII from medulla.



DRAW ZONE 2 -- Diagram a face showing the motor branches of CN VII. Then show Anna's right-sided palsy including the forehead. Compare with a UMN stroke that would spare the forehead.

Five facial nerve branches: temporal, zygomatic, buccal, mandibular, cervical. The temporal branch is what fails Anna's forehead.

RETURNING TO ANNA

One nerve, multiple branches, all the findings

Anna's case is a peripheral CN VII palsy — distinguished from a stroke by the forehead involvement.

"Why forehead is involved"

Peripheral lesion of CN VII. Takes out the entire nerve including the temporal branch to forehead muscles. UMN stroke would spare forehead.

"Eye won't close"

CN VII innervates orbicularis oculi (eye closure). Loss → lagophthalmos (can't close eye) → corneal exposure → ulceration risk. Critical to lubricate and tape closed at night.

"Loss of taste anterior 2/3 of tongue"

Chorda tympani branch of CN VII carries taste from anterior 2/3. Posterior 1/3 is CN IX (unaffected).

"Hyperacusis"

Loss of CN VII branch to stapedius. Stapedius normally dampens loud sounds by tensing the ossicles. Without it, sounds feel uncomfortably loud.

"Likely cause"

Bell's palsy: idiopathic, often associated with herpes simplex reactivation in the geniculate ganglion. Inflammation swells the nerve in the tight bony facial canal, leading to compressive demyelination.

"Prognosis"

~70% recover fully without treatment; oral corticosteroids started within 72 hours improve outcomes. Most recovery happens within 3-6 months. Some have residual weakness.

Forehead, taste, stapedius, lacrimation. Anna's exam is a CN VII anatomy lesson.

